

Transformations

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Which transformation moves every point of a shape the same distance in the same direction, without turning or resizing it?

- 1 The shape simply slides across the grid.
- 2 The image is congruent to the original — same shape, same size, same way up.

Answer: a translation

2 [2 marks]

Translate the point (3, 2) by the vector $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$

Write down the coordinates of the image.

- 1 The top number moves the point horizontally and the bottom number vertically.
- 2 $x: 3 + 4 = 7$
- 3 $y: 2 + (-5) = -3$

Answer: (7, -3)

3 [2 marks]

Reflect the point (3, 5)

(a) in the x -axis

(b) in the y -axis

- 1 **(a)** Reflecting in the x -axis changes the sign of the y -coordinate: (3, -5)
- 2 **(b)** Reflecting in the y -axis changes the sign of the x -coordinate: (-3, 5)

Answer: (a) (3, -5) (b) (-3, 5)

4 [2 marks]

Rotate the point (4, 0) through 90° anticlockwise about the origin.

- 1 The point is 4 units along the positive x -axis.
- 2 Turning 90° anticlockwise takes it to 4 units up the positive y -axis.

Answer: (0, 4)

5

[3 marks]

Reflect the point (2, 3)

(a) in the line $y = x$

(b) in the line $y = 1$

1 (a) Reflecting in $y = x$ swaps the coordinates: (3, 2)

2 (b) The point is $3 - 1 = 2$ units above the line $y = 1$.

3 The image must be 2 units below the line, at $y = 1 - 2 = -1$

4 The x-coordinate does not change, so the image is (2, -1).

Answer: (a) (3, 2) (b) (2, -1)

6

[3 marks]

(a) Enlarge the point (2, 3) by scale factor 3 about the origin.

(b) A shape is enlarged by scale factor 3. What happens to its area?

1 (a) Multiply both coordinates by 3: (6, 9)

2 (b) Both dimensions are 3 times as long, so the area is multiplied by 3^2 .

3 The area becomes 9 times as large.

Answer: (a) (6, 9) (b) the area is multiplied by 9

7

[3 marks]

Rotate the point (3, 1) about the origin

(a) through 180°

(b) through 90° clockwise

1 (a) A 180° rotation about the origin changes the sign of both coordinates: (-3, -1)

2 (b) For 90° clockwise, (x, y) maps to (y, -x).

3 So (3, 1) maps to (1, -3).

Answer: (a) (-3, -1) (b) (1, -3)

8

[3 marks]

Which of these transformations always give an image congruent to the original shape?

reflection · rotation · translation · enlargement

Explain your answer.

1 Congruent means exactly the same shape **and** the same size.

2 Reflections, rotations and translations move a shape without changing any lengths, so the image is congruent.

3 An enlargement changes the lengths (unless the scale factor is 1), so the image is a different size — similar but not congruent.

Answer: reflection, rotation and translation — enlargement changes the size

9

[4 marks]

Triangle P has vertices (1, 1), (3, 1) and (1, 4).

Triangle Q has vertices (1, -1), (3, -1) and (1, -4).

Describe fully the single transformation that maps P onto Q.

- 1 Compare each pair of points: the x -coordinates are unchanged and the y -coordinates have changed sign.
- 2 $(1, 1) \rightarrow (1, -1)$, $(3, 1) \rightarrow (3, -1)$, $(1, 4) \rightarrow (1, -4)$
- 3 Changing the sign of y only is a reflection in the x -axis.
- 4 A full description names the transformation and the mirror line.

Answer: a reflection in the x -axis (the line $y = 0$)

10

[4 marks]

Triangle T has vertices (1, 1), (2, 1) and (1, 3).

(a) Enlarge T by scale factor 2 about the origin. Write down the new vertices.

(b) Enlarge T by scale factor 2 about the point (1, 1). Write down the new vertices.

- 1 **(a)** About the origin, simply double both coordinates.
- 2 (2, 2), (4, 2) and (2, 6)
- 3 **(b)** About (1, 1), first write each point as a step from the centre, then double it.
- 4 (1, 1) is the centre, so it does not move.
- 5 (2, 1) is 1 right of the centre; doubling gives 2 right, so the image is (3, 1).
- 6 (1, 3) is 2 up from the centre; doubling gives 4 up, so the image is (1, 5).

Answer: (a) (2, 2), (4, 2), (2, 6) (b) (1, 1), (3, 1), (1, 5)

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Rotate the point (2, 5) through 90° clockwise about the origin.

- 1 For a 90° clockwise rotation about the origin, (x, y) maps to $(y, -x)$.
- 2 So (2, 5) maps to (5, -2).

Answer: (5, -2)

2

[2 marks]

Enlarge the point (4, -2) by scale factor -1 about the origin.

- 1 A negative scale factor puts the image on the opposite side of the centre.
- 2 Multiply both coordinates by -1: (-4, 2)

Answer: (-4, 2)

3

[3 marks]

(a) Enlarge the point (5, 3) by scale factor $\frac{1}{2}$ about the origin.

(b) A shape of area 40 cm^2 is enlarged by the same scale factor. Work out the new area.

1 (a) Halve both coordinates: (2.5, 1.5)

2 (b) Area scale factor = $(\frac{1}{2})^2 = \frac{1}{4}$

3 $40 \times \frac{1}{4} = 10 \text{ cm}^2$

Answer: (a) (2.5, 1.5) (b) 10 cm^2

4

[3 marks]

Triangle A has vertices (0, 0), (2, 0) and (0, 3).

Triangle B has vertices (0, 0), (-2, 0) and (0, -3).

Describe fully the single transformation that maps A onto B.

1 Both coordinates have changed sign in every case.

2 That is exactly what a rotation of 180° about the origin does.

3 The shape is the same size, so it is not an ordinary enlargement.

Answer: a rotation of 180° about the origin

5

[4 marks]

A shape is reflected in the x -axis, and the image is then reflected in the y -axis.

Show that the result is the same as a single rotation, and describe that rotation fully.

1 Start with a general point (a, b).

2 Reflecting in the x -axis gives (a, -b).

3 Reflecting that in the y -axis gives (-a, -b).

4 So the combined effect changes the sign of both coordinates.

5 That is a rotation of 180° about the origin, for every point of the shape.

Answer: (a, b) $\rightarrow$ (-a, -b), which is a rotation of 180° about the origin

6

[4 marks]

Enlarge the point (4, 3) by scale factor -2 about the centre (1, 1).

1 Write the point as a step from the centre: $(4 - 1, 3 - 1) = (3, 2)$

2 Multiply that step by the scale factor -2: (-6, -4)

3 The negative sign sends the image to the opposite side of the centre.

4 Add the step back on to the centre: $(1 - 6, 1 - 4)$

5 The image is (-5, -3).

Answer: (-5, -3)

7

[4 marks]

(a) A shape is reflected in the line $x = 3$. Which points of the shape do not move? Explain.

(b) A shape is rotated about the point $(2, 5)$. Which points do not move? Explain.

1 Points that do not move under a transformation are called invariant points.

2 (a) Any point already on the mirror line reflects onto itself.

3 So the invariant points are those with $x = 3$.

4 (b) A rotation turns every point around the centre.

5 Only the centre itself stays where it is, so $(2, 5)$ is the only invariant point.

Answer: (a) the points on the line $x = 3$ (b) only the centre $(2, 5)$

8

[4 marks]

Describe fully the single transformation equivalent to each pair.

(a) A translation by $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$ followed by a translation by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$

(b) A reflection in $y = x$ followed by a reflection in $y = -x$

1 (a) Add the two vectors: $5 + (-2) = 3$ and $0 + 3 = 3$

2 So the single transformation is a translation by $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$

3 (b) Take a general point (a, b) . Reflecting in $y = x$ swaps the coordinates to give (b, a) .

4 Reflecting (x, y) in $y = -x$ gives $(-y, -x)$, so (b, a) becomes $(-a, -b)$.

5 Both coordinates have changed sign, which is a rotation of 180° about the origin.

Answer: (a) a translation by $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$ (b) a rotation of 180° about the origin

9

[4 marks]

Rotate the point $(5, 2)$ through 90° anticlockwise about the centre $(1, 1)$.

1 Write the point as a step from the centre: $(5 - 1, 2 - 1) = (4, 1)$

2 For a 90° anticlockwise rotation, a step (x, y) becomes $(-y, x)$.

3 So $(4, 1)$ becomes $(-1, 4)$.

4 Add the new step back on to the centre: $(1 - 1, 1 + 4)$

5 The image is $(0, 5)$.

Answer: $(0, 5)$

Triangle T has vertices (1, 1), (4, 1) and (1, 3).

(a) T is enlarged by scale factor -2 about the origin to give T' . Find the vertices of T' .

(b) Work out the area of T.

(c) Work out the area of T' .

1 (a) With the origin as centre, multiply every coordinate by -2 .

2 (1, 1) $\rightarrow$ (-2, -2), (4, 1) $\rightarrow$ (-8, -2), (1, 3) $\rightarrow$ (-2, -6)

3 (b) T is right-angled: the base from (1, 1) to (4, 1) is 3 and the height from (1, 1) to (1, 3) is 2.

4 Area = $\frac{1}{2} \times 3 \times 2 = 3$ square units

5 (c) The area scale factor is the square of the length scale factor, and squaring removes the minus sign: $(-2)^2 = 4$

6 Area of $T' = 3 \times 4 = 12$ square units

Answer: (a) (-2, -2), (-8, -2), (-2, -6) (b) 3 square units (c) 12 square units